

1. Given a machine code, what do we understand by looking at the Opcode field? [2]
2. Suppose you have converted LD X9, 10(X21) to its equivalent RISC-V machine code. Now, based on which field, the machine will understand the size and extension type of the data being loaded into register X9? [3]
3. Given the following 32-bit binary machine codes, determine the corresponding RISC-V assembly instruction. [2 + 2]

Use the provided opcode and funct reference table for identification

- a. 0000000 01011 01010 000 01001 0110011
- b. 0000000 00101 10001 101 10010 0010011

Instruction	Opcode	Funct3	Funct7
add	0110011	000	0000000
xor	0110011	100	0000000
ld	0000011	011	-
srli	0010011	101	0000000
sub	0110011	000	0100000

4. Consider the following code:

```
int hishab(int i, int j) {
    i = i + 8;
    j = j - a[a [0]];
    while (j < i) {
        j = a [i];
        i += 1;
    }
    return j;
}

k = 2;
l = 3;
a [0] = hishab(k, l);
```

- a. Construct the equivalent RISC-V code of the above-mentioned C code. Assume that the variables k and l are in registers X5 and X6 respectively. a is a word array, which has its base address stored in X18. [15]
- b. From your converted codes, pick any instruction from each of the three formats(R, I, S) and convert them into their corresponding binary values. [6]

Note: For opcode, funct3 or any other field values that you do not know just put "X".